

Vojtěch Špaček

Links: [GitHub](#) | [LinkedIn](#) | [X/Twitter](#) **Contact:** sparesparrow@protonmail.ch | +420703454180 | [Schedule a call](#)

AI/LLM & embedded software engineer — safety-critical verification, hardware-in-the-loop test automation, Model Context Protocol tooling

Personal Details

- **Nationality:** Czech Republic (EU citizen) · **Work permit:** none required in the EU/EEA; **B-permit eligible** for Switzerland
- **Residence:** Brno, Czech Republic · willing to relocate or commute
- **Languages:** Czech (native) · Slovak (native) · English (C1) · German (A1, basic)

Profile

- Senior engineer in **safety-critical verification and embedded software**. Sole owner of the hardware-in-the-loop verification suite and automated bench for a safety-certified two-stage gas furnace ignition controller (UL/CSA, ANSI Z21.20, NFPA 54), closing a ~148-case gap where stimulus windows had never been measured.
- A decade across **aerospace** (Honeywell onboard diagnostics), **scientific instruments** (Thermo Fisher electron microscopy) and **network security** (Kernun inspection proxy and firewall).
- Brings **agentic-AI tooling (Model Context Protocol)** into regulated engineering workflows.

Core Skills

- **C++** (*Senior*, C++17/20) · **Python** (*Senior*) · **C#** (*Medior*) · **JavaScript** (*Medior*) · **Rust**, **Qt/QML** (*working knowledge*)
- Safety-certified product verification, requirement traceability, test-evidence reporting
- Hardware-in-the-loop benches, [pyvisa/USBTC/SCPI](#), [CAN](#), [Modbus](#), deterministic multi-threaded device ownership
- Embedded & RTOS, ARM, state-machine design, HAL & IPC, [gRPC](#), Flatbuffers, Conan, CI/CD
- Linux, [Netlink](#), [nftables](#), TLS 1.3 inspection, Kerberos/LDAP/OCSP

Professional Experience

Resideo (Ademco) - CoolPeople Contractor

Test Automation Engineer (Python, C#) (01/2026 – 09/2026): Hardware-in-the-loop verification of a safety-certified two-stage gas furnace ignition controller ([UL/CSA](#), [ANSI Z21.20](#), [NFPA 54](#)), as sole owner of the verification test suite and of the automated bench that drives it.

- Designed an [Arrange/Act/Assert](#) validation pipeline in [Python](#) with eight independent gates covering preconditions, stimulus timing, state sequence, alarms, signal history, cross-layer consistency, timer accuracy and requirement traceability.
- Built a data-driven generator expanding a [YAML](#) catalogue of 270+ state transitions, 38 states and 33 events into roughly 290 executable tests, removing hand-written duplication.
- Engineered the bench device layer so each instrument is owned by a single thread against a shared monotonic clock, covering [pyvisa/USBTC/SCPI](#) programmable AC source, [CAN](#) serial-bus monitoring and [Modbus](#) relay and probe I/O.

Thermo Fisher Scientific - Ixperta Contractor

Software Developer (C++): Electron Microscopy software development for high-resolution detector hardware operation and image acquisition.

- Implemented server-side software in **C++**, including **hardware abstraction layers (HAL)** and **inter-process communication (IPC)** using **COM** and **MIDL** on Windows.
- Developed embedded software for **FPGA** hardware, utilizing network communication protocols (**gRPC**) in C++, Python, and Linux environments.

Honeywell

- **Project**: Onboard Maintenance System that monitors the condition of aircraft and diagnoses issues during flight.
- Architected and implemented **DiagTest framework**, a sophisticated **state machine-based diagnostic test engine** in **C++** managing 15+ operational states for aircraft member system testing during flight and maintenance operations.

Trusted Network Solutions

- **Implemented a forward proxy** using **C++** on **Linux**, featuring **HTTP/HTTPS proxy with TLS/SSL inspection**, **OCSP integration** for certificate validation, **Kerberos authentication**, and **LDAP integration** for user/group lookup.
- **Developed and maintained ipmon**, a sophisticated network monitoring solution demonstrating deep expertise in:

IBM — Automation Developer / IT Systems Specialist

- Automation application replacing manual ticketing and server-administration workload (**Bash**, **PowerShell**, **JavaScript**); monitored Unix/Windows production servers.

Education & Training

- **Masaryk University, Faculty of Informatics** — Computer Science coursework, *no degree awarded*. Programming, Linux, Networking.
- **Coursera / DeepLearning.AI** — Deep Learning Specialization (Andrew Ng), data analysis, prompt engineering, building LLM agents.
- **Self-directed** — design patterns and software architecture; Hack The Box; Certified Ethical Hacker syllabus (exam not taken).

Selected Open Source

- **MCP-Prompts** (97 ★, 21 forks) — Model Context Protocol server for prompt templates and LLM interactions. On npm, Docker Hub and the MCP directories; Rust port on crates.io.
- **rust-network-mgr** — Rust daemon driving **nftables** rules via **Netlink**; lock-free, epoll-based. Reimplementation of my C++20 **ipmon** daemon. On crates.io.

References

Available on request — two to three referees and Arbeitszeugnisse.